A Horizon-Centered Formalization of Cosmological Dynamics and Perturbations

Kevin Muñoz

Abstract

We develop a horizon-centered formalization of cosmological dynamics and perturbations in which the apparent horizon is treated as the primary physical degree of freedom. Building on a previously introduced conceptual framework, we show that standard Friedmann dynamics can be recovered from horizon thermodynamics and that cosmological perturbations may be consistently described as fluctuations of the causal boundary.

We derive a closed linear perturbation pipeline linking horizon fluctuations to density perturbations and to the comoving curvature perturbation. Under minimal statistical assumptions, the framework reproduces a nearly scale-invariant spectrum compatible with observations. The approach remains fully consistent with General Relativity and standard phenomenology, while providing a boundary-based interpretation of cosmological evolution.

1 Introduction

Standard cosmology describes the universe in terms of bulk spacetime dynamics governed by General Relativity, with expansion encoded in the scale factor $a(t)$. Horizons are typically treated as derived structures.

In contrast, horizon thermodynamics suggests that causal boundaries encode physically relevant information, with entropy scaling with area rather than volume [1–3]. This motivates alternative perspectives in which causal horizons play a structurally primary role.

The present work provides a formal realization of a horizon-centered framework in which cosmological dynamics and perturbations are described in terms of the apparent horizon. The goal is not to modify General Relativity, but to reformulate its cosmological content in terms of causal accessibility and boundary thermodynamics.

2 Background Geometry and Horizon Variable

We consider a homogeneous and isotropic spacetime with Hubble parameter:

$$H = \frac{\dot{a}}{a} \tag{1}$$

The apparent horizon radius is:

$$R_A = \frac{1}{\sqrt{H^2 + \frac{k}{a^2}}} \tag{2}$$

For spatially flat geometry ($k = 0$):

$$R_A = \frac{1}{H} \tag{3}$$

We take $R_A(t)$ as the fundamental variable, interpreting cosmological evolution as the growth of a causally accessible domain.

3 Horizon Thermodynamics and Background Dynamics

We assign thermodynamic quantities to the horizon:

$$S = \frac{A}{4G} = \frac{\pi R_A^2}{G} \tag{4}$$

$$T = \frac{1}{2\pi R_A} \tag{5}$$

$$V = \frac{4}{3}\pi R_A^3 \tag{6}$$

$$E = \rho V \tag{7}$$

We impose the first law:

$$dE = TdS - PdV \tag{8}$$

Using energy conservation:

$$\dot{\rho} + 3H(\rho + P) = 0 \tag{9}$$

and $R_A = 1/H$, we obtain:

$$\dot{H} = -4\pi G(\rho + P) \tag{10}$$

which integrates to:

$$H^2 = \frac{8\pi G}{3}\rho + C \tag{11}$$

Thus, standard Friedmann dynamics are recovered from horizon thermodynamics.

4 Perturbations of the Horizon

We introduce perturbations in the horizon radius:

$$R_A(t, \mathbf{x}) = \bar{R}_A(t)\,[1 + \psi(t, \mathbf{x})] \tag{12}$$

where:

$$\psi = \frac{\delta R_A}{R_A} \tag{13}$$

is dimensionless and treated as the fundamental perturbation variable.

5 Relation to Expansion and Density Perturbations

Using $H = 1/R_A$, we obtain at linear order:

$$\frac{\delta H}{H} = -\psi \tag{14}$$

From the Friedmann equation:

$$H^2 \propto \rho \tag{15}$$

we obtain:

$$\frac{\delta\rho}{\rho} = 2\frac{\delta H}{H} = -2\psi \tag{16}$$

Thus:

$$\delta\rho \propto \psi \tag{17}$$

6 Relation to the Curvature Perturbation

We identify the comoving curvature perturbation as:

$$\mathcal{R} \sim \psi \tag{18}$$

This identification reflects that ψ directly controls local expansion. A complete gauge-invariant treatment remains an open problem [4].

7 Statistical Description

We assume Gaussian statistics:

$$\langle \psi(\mathbf{k})\psi(\mathbf{k}') \rangle = (2\pi)^3 \delta(\mathbf{k} + \mathbf{k}') P_\psi(k) \tag{19}$$

with:

$$P_\psi(k) = A k^{n_s - 1} \tag{20}$$

Then:

$$P_{\mathcal{R}}(k) = P_\psi(k) \tag{21}$$

8 Effective Horizon Field

We introduce an effective scalar field defined on the horizon:

$$\phi(\Omega, t) \tag{22}$$

with action:

$$S = \int dt\, d\Omega\, R_A^2 \left[(\partial_t \phi)^2 + \frac{c_s^2}{R_A^2} (\nabla_\Omega \phi)^2 + m_{\text{eff}}^2 \phi^2 \right] \tag{23}$$

and identify:

$$\psi \sim \phi \tag{24}$$

9 Mode Structure and Freeze-out

Expanding in spherical harmonics:

$$\phi = \sum_{l,m} \phi_{lm}(t) Y_{lm}(\Omega) \tag{25}$$

the mode frequency is:

$$\omega_l^2 = \frac{c_s^2}{R_A^2} l(l+1) + m_{\text{eff}}^2 \tag{26}$$

Modes freeze when:

$$\omega \sim H \tag{27}$$

Using $R_A = 1/H$, this yields:

$$l \sim k R_A \tag{28}$$

10 Spectrum

Assuming scale-invariant correlations:

$$\langle \phi(x)\phi(0)\rangle \sim \frac{1}{|x|^{\alpha}} \tag{29}$$

we obtain:

$$P(k) \sim k^{\alpha-3} \tag{30}$$

For $\alpha \approx 3$:

$$n_s \approx 1 \tag{31}$$

Introducing:

$$m_{\text{eff}}^2 = \beta H^2 \tag{32}$$

yields:

$$n_s - 1 \approx -2\beta \tag{33}$$

11 Connection to Observables

The angular power spectrum is given by:

$$C_\ell = 4\pi \int \frac{dk}{k} P_{\mathcal{R}}(k)|\Delta_\ell(k)|^2 \tag{34}$$

with:

$$\ell \sim kr_* \tag{35}$$

12 Interpretation

The framework provides a dual description:

- Standard: bulk fields generate perturbations
- Horizon-centered: boundary fluctuations encode perturbations

13 Limitations and Open Problems

- No microscopic model of horizon degrees of freedom
- Incomplete gauge-invariant formulation of $\mathcal{R}$
- Mapping from boundary to bulk not fully derived
- Origin of effective horizon field remains unspecified

14 Conclusion

We have constructed a horizon-centered formalization of cosmological dynamics and perturbations that reproduces standard results while providing a boundary-based interpretation. The framework is compatible with General Relativity and observational data, and identifies a clear set of open problems for future development.

References

[1] J. D. Bekenstein, "Black holes and entropy," Phys. Rev. D 7, 2333 (1973).

[2] S. W. Hawking, "Particle creation by black holes," Commun. Math. Phys. 43, 199 (1975).

[3] G. W. Gibbons and S. W. Hawking, "Cosmological event horizons, thermodynamics, and particle creation," Phys. Rev. D 15, 2738 (1977).

[4] V. Mukhanov, "Physical Foundations of Cosmology," Cambridge University Press (2005).